AI-powered book review summarization in Norwegian

Candidate 108, Candidate 114, Candidate 122

Abstract

In recent years, the volume of text corpora, including user-generated online book reviews, has been growing rapidly. Observing this, we developed a system that took Norwegian book reviews and discussions collected from Bokelskere.no and generated abstractive, sentiment-aware summarized reviews. We first applied the encoder model Norbert3-large_TSA to perform a targeted sentiment analysis on the reviews. Norbert3-large_TSA extracted sentiment BIO-tags from the reviews to construct the summarization input. For the summarization task, we employed two pretrained Norwegian language models, Llama-2-7b-chat-norwegian and NorMistral-7b-warm. To assess the quality of the summarized reviews, we performed a human evaluation following the SummEval framework. The results showed that Llama-2-7b-chat-norwegian outperformed NorMistral-7b-warm in the summarization task for the 40 abstractive reviews. Future work should explore various models and fine-tuning strategies to improve human evaluation scores, coherence, and understandability, as well as incorporate automatic metrics like ROUGE.

1 Introduction

As the volume of available text corpora, including online reviews, grows quickly, text summarization is needed to reduce information overload (Mohd et al., 2023). Text summarization is the process of reducing the size of a document while maintaining meaning. This is among the most researched fields within Natural Language Processing (NLP) (Gupta and Gupta, 2019), with review summarization receiving increasing attention. Summarization techniques are typically categorized into extractive and abstractive approaches. Extractive text summarization methods identify the most meaningful phrases based on statistical features. Abstractive methods consist of generating novel sentences that capture the key ideas and essence of the source text, usually through paraphrases or new words. The development of transformer-based models like BART (Lewis et al., 2019) and transformer-based language models (Raffel et al., 2023) has shifted the attention more to abstractive techniques and hybrid methods (Mehta, 2016; Jurafsky and Martin, 2025). Furthermore, methods incorporating sentiment signals from the text in the summarized review have been presented, as empirical experiments show that including sentiments can outperform state-of-the-art approaches in various metrics (Yang et al., 2018; Han et al., 2022).

As experiments by Radford et al. (2019) and Raffel et al. (2023) have shown that pretrained large language models can learn to do abstractive summarization, we present a novel approach using causal autoregressive language models to enhance abstractive summarization with fine-grained sentiment BIO-tags extracted from user-generated book reviews. To our knowledge, there has not been any summarization work on book reviews in the Norwegian language employing sentiment BIO-tags. The research questions we address are defined as follows:

RQ1: How can we use abstractive summarization to create Norwegian book review summaries?

RQ2: How do the models Llama-2-7b-chat-norwegian and NorMistral-7b-warm compare in generating Norwegian sentiment-aware abstractive summaries?

We will achieve this by building a scalable summarization pipeline where user reviews and discussions are processed by a sentiment tagging language model that assigns sentiment targets as either "targ-Positive" or "targ-Negative". These BIO-tags are then combined to form the input feed to our two transformer-based models that output a single coherent summary of the user reviews for each book. The description of the models, datasets used for fine-tuning and summary generation will be found in Sections 2 and 3. In section 4, we perform an evaluation of the models that will be performed through human evaluation using SummEval metrics. The results indicate that Llama-2-7b-chat-norwegian outperforms the Normistral-7b-warm model in three out of four dimensions. Section 5 describes the error analysis. The following Section 6 describes the challenges and limitations that we experienced throughout the project. Finally, in Section 7 we outline future directions for fine-tuning and the incorporation of automatic evaluation metrics such as ROUGE.

1.1 Related Work

Abstractive summarization Previous studies on abstractive summarization have focused extensively on news articles, mainly due to the availability of large benchmark datasets (Nallapati et al., 2016; Zhou et al., 2017; Paulus et al., 2017) and, on a smaller scale, scientific documents (Zakkas et al., 2024; Ulker and Ozer, 2024). In recent years, research has increasingly examined user-generated content, including reviews. For example, Angelidis and Lapata (2018) introduces a method for summarizing opinions using two weakly supervised components that identify these opinions and generate summaries from multiple reviews. Similarly, Xu et al. (2023) proposes an approach for generating personalized review summaries that tailor the output based on user preferences.

Sentiment-aware Summarization In a recent work, Han et al. (2022) proposed a scalable methodology for abstractive opinion summarization from online reviews utilizing aspect terms and sentiment polarities. Likewise, Yang et al. (2018) devised a neural attentive model that generates aspect- and sentiment-aware abstractive summaries of given reviews.

Pre-trained Language Models and Norwegian Resources State-of-the-art summarization methods today are driven by transformer-based language models (Zhang et al., 2020; Radford et al., 2019; Wang et al., 2023). To support Norwegian, various Norwegian pretrained models have emerged (Samuel et al., 2025; Liu et al., 2024) however, a majority of research on Norwegian text summarization has used news article corpora as its primary evaluation benchmark (Touileb et al., 2025; Mikhailov et al., 2025). Furthermore, Samuel et al. (2023) introduced a comprehensive suite of benchmark tasks for Norwegian language models, allowing systematic evaluation and comparison.

Contribution of our project Limited research has examined the integration of sentiment BIO-tags in abstractive news summarization, particularly using Norwegian text data. This project differs by using BIO-tags for Norwegian user-generated review summarization. We contribute to the growing field of Norwegian text summarization research by experimenting with a book review corpus.

2 Dataset

The dataset used in this project is the corpus of user-generated book reviews and discussions from Bokelskere.no. The corpus can be downloaded from the website of the National Library of Norway[1]. It was last updated in 2020, but the data is from 2019 and is structured in a JSON format, with each object representing a review or comment. It has 10 columns and 218,937 rows, where each object has a `text` column that represents a review. The reviews are written primarily in Scandinavian languages, such as Swedish, Norwegian Bokmål, and Nynorsk. The total number of word tokens in the `text` column is 1.5 million. All reviews are rated with a numerical score on a scale from 1 to 6. In the project, we will only work with a subset of the 40 most reviewed books and their corresponding user reviews. Each review has a creation date, user ID, and post title. A complete overview of the dataset and metadata fields is reported in tables 1 and 2.

Property	**Value**	**Notes**
# Posts	219,000	User reviews/comments
Total tokens	1,500,000	In the “text” field
Avg. review length	72.4 words	Mean
Time span	2010–2019	Date field
Languages	Bokmål/Nynorsk	Mixed

Table 1: Overview of the Bokelskere.no corpus.

2.1 Data preprocessing

For data preprocessing, we first removed user reviews that were not in a Scandinavian language, missing a book title, author, and score, as these fields are crucial for our summarization task. The remaining texts were cleaned by stripping URLs, digits, and non-standard symbols beyond punctuation.

[1] https://www.nb.no/sprakbanken/en/resource-catalogue/oai-nb-no-sbr-53/

Field	Description
post_id	unique identifier for the review
date	date of posting
user_id	unique identifier for the reviewer
isbn13	ISBN-13 of the reviewed book
post_title	title of the review
text	full review text
score	rating (1–6)
main_title	title of the book
author	author of the book
parent_id	ID of a parent review if this is a comment

Table 2: JSON metadata fields in the Bokelskere.no corpus.

2.2 Sentiment BIO-Tagging

We utilized the publicly released Targeted Sentiment Analysis (TSA) model `ltg/norbert3-large_tsa`, trained on the `ltg/norec_tsa`[2] dataset derived from the `Norec_fine` (Samuel et al., 2023). Each token in `norec_tsa` is labeled as either `targ-Positive` or `targ-Negative`. The checkpoint achieves an F1-score of 89.3% on its held-out test split. After conducting the TSA model on our cleaned reviews, we identified 1508 tokens (75.2%) as `targ-Positive` and 497 tokens (24.8%) as `targ-Negative`, indicating a strong positive skew in the input to our summarization models. Additionally, we created columns in our dataframe holding the positive and negative tokens, labeled `positive_snippets` and `negative_snippets`, and the counts for these tokens.

2.3 Summarization input

We grouped all reviews by title, preserved the first author, and average mean score. All associated raw user reviews were concatenated into one string. Then we flattened each title's `positive_snippets` and `negative_snippets`, removed duplicates, and removed snippets with less than two words. Finally, for each title, we built a single summarization input by prefixing positive snippets with "Leserne likte:" and negative snippets with "Leserne mislikte:", separated by a comma and punctuated with a period.

3 Model selection and methods

For the targeted sentiment analysis (TSA) task, we used `NorBert3-Large_TSA` (Samuel et al., 2023). For the summarization task, we selected `RuterNorway/Llama-2-7b-chat-norwegian` (Touvron et al., 2023) and `norallm/normistral-7b-warm` (Samuel et al., 2025).

[2]https://huggingface.co/datasets/ltg/norec_tsa

NorBert-3-Large_TSA NorBert3-Large_TSA is a fine-tuned version of Norbert3-large that identifies the sentiment target of an opinion and classifies whether the target is positive or negative by sequence labeling.

Llama-2-7b-chat-norwegian Llama-2-7b-chat-norwegian is a Norwegian fine-tuned version of Meta's open-source large language model Llama-2-7b-chat (Touvron et al., 2023). The language model comes with 7 billion and 13 billion parameters.

NorMistral-7b-warm NorMistral-7b-warm is a competitive large language model pretrained on 260 billion subword tokens, which achieves robust performance in a wide range of Norwegian NLP tasks, including abstractive summarization (Samuel et al., 2025; Hugging Face, 2025).

3.1 Fine-tuning setup

In order to reduce memory usage and address computational constraints, we combined a 4-bit quantized Low Rank Adaptation (QLoRA) configuration with supervised fine-tuning (Hou et al., 2024). We chose this approach because it significantly reduces the number of parameters and training requirements while preserving model performance in terms of inference and latency.

3.1.1 Fine-tuning dataset

The Norwegian Summarisation (NorSumm) Benchmark Dataset (Touileb et al., 2025) dataset has 378 human-authored summaries of notable Norwegian news outlets across different domains. Initially, the dataset had only validation and test splits, each containing 30 validation and 33 test samples for both Norwegian Bokmål and Nynorsk subsets. To enable training, we combined these validation subsets, resulting in 60 examples. This combined dataset was then sectioned into training and validation subsets using an 80-20 split, resulting in 48 training and 12 validation instances. We then merged the original test subsets for Bokmål and Nynorsk and retained them as the final test set. Table 3 describes the number of samples and the average article and summary lengths, in words, for each split and Norwegian variant.

Split	#Ex.	Avg. Art.	Avg. Sum.
Val.	30	585.1	285.7
Test	33	792.0	263.0

(a) Bokmål

Split	#Ex.	Avg. Art.	Avg. Sum.
Val.	30	585.1	288.1
Test	33	792.0	274.8

(b) Nynorsk

Table 3: NorSumm dataset statistics (number of examples, average article/summary length in words).

3.1.2 Training details

The fine-tuning implementation was based on the Colab notebook by Labonne (2024)[3]. It was conducted over the training split for 4 epochs and 74 update steps on a single NVIDIA A100 GPU. Model checkpoints were saved at the end of each epoch, and the final model was chosen as the one achieving the highest ROUGE-L on the held-out validation set. During training, we followed a similar setup presented by Hou et al. (2024) to fine-tune smaller datasets. We had a learning rate of 1×10^{-5}, four epochs with a paged AdamW-8bit optimizer from BitsAndBytes ($\beta_1 = 0.9$, $\beta_2 = 0.999$,$\epsilon = 1\times10^{-8}$). We applied QLoRA to the models' projection layers using a rank $r = 32$, scaling $\alpha = 64$, and dropout rate of 0.05 to efficiently adapt the 7B LLaMA with minimal extra parameters. A summary of the hyperparameters used during training can be found in Table 4. Results of the fine-tuning can be seen in Table 5.

Hyperparameter	Value
Learning rate	1×10^{-5}
Optimizer	paged_adamw_8bit ($\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 1 \times 10^{-8}$)
Batch size / Grad. accu.	4 seq./device, 2 steps (eff. batch 8)
LR scheduler	constant
Epochs	4

Table 4: Training hyperparameters for fine-tuning Llama-2-7b-chat-norwegian and NorMistral-7b-warm on NorSumm.

[3]Labonne, Maxime. "Fine-Tune Your Own LLaMA 2 Model in a Colab Notebook." 2024. https://colab.research.google.com/github/ashishpatel26/LLM-Finetuning/blob/main/2.Fine_Tune_Your_Own_Llama_2_Model_in_a_Colab_Notebook.ipynb

Metric	Llama-2-7b-chat-norwegian	NorMistral-7b-warm
Validation loss	2.37	2.23
ROUGE-1	41.95	57.00
ROUGE-2	9.80	20.90
ROUGE-L	29.58	42.04
ROUGE-L-sum	40.84	55.56

Table 5: Evaluation metrics for the fine-tuned Llama-2-7b-chat-norwegian and NorMistral-7b-warm models on the NorSumm test set.

3.2 Summarization Prompt and generation parameters

A summarization prompt consisting of instructions and input was used to generate qualitative and abstractive summaries of the sentiment tags. For both models, we used the following prompt described below based on the templates described in the model cards found on Hugging Face:

```
prompt = (
    "### Instruksjon:\n"
    "Du er en
        bokanmeldelsessammendragsgenerator
        . Gjør om følgende 'Leserne
        likte'- og 'Leserne mislikte'-
        tagger til et 3-5 setningers,
        fullstendig abstrakt sammendrag
        på norsk av hvordan boken ble
        mottatt. Ikke gjengi taggene
        ordrett - skriv flytende og
        sammenhengende prosa på norsk.\n
        \n"
    "### Inndata:\n"
    f"{text}\n\n"
    "### Sammendrag:"
)
```

For model inference, identical to Liu and Demberg (2024), we used a beam search size of 4, a penalty length of 3.0, and set a no-repeat n-gram of 3 on both models to ensure a balance between computational costs and outputs that were consistent.

4 Evaluation and results

4.1 Human evaluation

In the absence of gold standard summaries in our dataset, we conducted a manual evaluation. To perform our human evaluation, we first decided on a 1-3 Likert scale to individually rate two generated summaries on the four dimensions presented by Fabbri et al. (2021). Then, we compared our results on each dimension for each generated summary. Here, we defined the criteria and the meaning of the Likert scale and then proceeded to individually rate three more generated summaries based on our criteria. After this, Fleiss' Kappa for inter-annotator agreement was calculated, and the scores for this can be seen in Table 6. To guarantee reliable annotations, we set the Fleiss' Kappa threshold to at least 0.60 for all evaluation dimensions (Wong et al., 2021). As the scores were adequate, we assigned the remaining 35 generated summaries between us.

Measure	Coherence	Consistency	Fluency	Relevance
Fleiss' κ	0.659	0.700	0.789	0.605

Table 6: Inter-annotator agreement (Fleiss' κ) for each evaluation dimension after evaluating five books per rater.

SummEval To evaluate the summary, we used a set of criteria based on the summarization evaluations presented by Fabbri et al. (2021):

- **Coherence**: The overall quality, structure, and organization of all sentences in the generated summarized reviews.
- **Consistency**: Alignment between the summary and the sourced reviews, ensuring that the summary consists only of factual statements that can be derived from the source document.
- **Fluency**: Quality of individual sentences, focusing especially on correct syntax to secure the reading flow of the summaries.
- **Relevance**: Selection of significant content from the user-generated reviews.

4.2 Results

The results from evaluating Llama-2-7b-chat-norwegian and NorMistral-7b-warm on the 40 summarized Norwegian book reviews using the four evaluation dimensions for human evaluation are shown in Table 7.

Dimension	Llama-2-7b-chat-norwegian	NorMistral-7b-warm
Coherence	**2.18**	2.05
Consistency	**1.75**	1.43
Fluency	2.10	**2.25**
Relevance	**2.15**	1.40

Table 7: Average human evaluation scores (1–3) for each SummEval dimension and each model.

Llama-2-7b-chat-norwegian Results showed that the model repeatedly outperformed norMistral-7b-warm in coherence, consistency, and relevance. Indicating that the summaries generated were generally well-structured, aligned with information derived from the source reviews, and preserved the essence of the reviews. More specifically, the model outscored NorMistral-7b-warm by 0.13 for coherence, 0.32 for consistency, and 0.75 for relevance. The higher consistency and relevance score than the NorMistral-7b-warm model indicates that this model is more faithful to the input source and its relevant parts. This might make the summarized reviews more usable and trustworthy. An example of a summary scoring three points for each dimension is shown in Table 8.

NorMistral-7b-warm The model displayed above-average performance in fluency, outscoring the Llama-2-7b-chat-norwegian model by 0.15 points. This performance reflected the model's diverse language, grammatical correctness, and linguistic flow in the generated summaries. Likewise, an example of the solid fluency of the model is shown in Table 8. The model scored above average on coherence, enhancing that the grammatical and syntactical aspect of summaries generated is a strength of the NorMistral-7b-warm model. These advantages indicate that the summarized reviews are readable and have a logical progression, which can make them more accessible.

Model	Example Summary
Llama-2-7b	Boka ble generelt møtt med positiv reaksjon fra leserne, som likte handlingen, historien og skildringen. De likte også språket og fortellerstilen, og mente at boka var en god fortelling av en familiehistorie. Imidlertid var noen lesere misfornøyde med bokens handling, som de mente ikke var tilstrekkelig spennende eller interessant.
NorMistral-7B	Lars Myttings bok om ved ble godt mottatt av leserne. Det ble også Svøm Med De Som Drukner som er en roman om familiehemmeligheter, om mysterier og om en mann som blir drevet til det ytterste av sin egen besettelse for å finne sannheten om sin egen fars død. Boken er en blanding av en familiehistorie, et mysterium og en kjærlighetshistorie. Den er skrevet på en måte som gjør at det er lett å leve seg inn i den, og det er vanskelig å legge den fra seg når man først har begynt å lese den.

Table 8: High-quality example summaries from Llama-2-7b-chat-norwegian and NorMistral-7b-warm.

5 Error analysis

We identified several errors during the human evaluation of the review summaries. In this section, we give a summary of the errors that commonly appeared. Some examples of the errors described can be seen in Table 9.

5.1 Errors related to coherence

The collective quality of the generated summaries was, for the most part, well maintained. We did observe some cases of unnecessary repetition in the summary. This was especially evident for the title "Fugletribunalet" with the summary containing the similar sentences "møtt med blandede reaksjoner blant leserne." and "Boka fikk en blandet mottakelse".

5.2 Errors related to consistency

Error analysis revealed that the generated summaries often included wrong information, such as the wrong title, characters, plots, and misinterpreting the main characters mentioned in the book as authors, and often stopping mid-sentence. For some specific books, such as "Politi", the models failed to output any summary. For the book "Jeg er Zlatan", the generated summary changed the title to "Ikke jeg som du tror jeg å være" which, after an internet search, is not a real book title, and the generated summary of the book "1Q84" contains the wrong author name.

5.3 Errors related to fluency

While fluency was generally well maintained, some model-generated summaries contained grammatical errors, including incomplete sentences, spacing, and punctuation errors. Additionally, a couple of summaries contained nonsensical words, like "forra" and "keplerkompletterer", in the sentence: "Boka 'Kepler' av Lars Keplerkompletterer de to forra bokene i serien, 'Joona' og 'Ildvittnet'", and stopped mid-word. While not a conclusive finding, these grammatical errors could arise due to imperfect training data.

5.4 Errors related to relevance

Some of the summaries were a copy-paste of the input, others described only the main characters and surroundings of the books rather than the sentiment tags, and at times they instead summarized or explained the step-by-step process of summarizing book reviews; this was seen for titles "Sandmannen" and "Øya". Examining the error cases where the models did not output anything relevant, we see that longer input led to issues with generating. Shortening the input on these error cases produced summarizations, though that often introduced new issues with the model hallucinating.

Error Type	Description
Coherence	Unnecessary repetition within the summary, leading to redundant phrasing.
Consistency	Incorrect or fabricated details (wrong titles, authors, characters, plots), hallucinations, and often truncated or missing summaries.
Fluency	Grammatical issues including incomplete sentences, spacing/punctuation errors, and nonsensical word fragments.
Relevance	Off-topic outputs (copy-paste of input, process descriptions), summaries focusing on the wrong aspects (e.g., characters rather than sentiment), or failure to generate when inputs are long.

Table 9: Summary of common error types observed in model-generated summaries.

6 Limitations and challenges

6.1 Limitations

Time and computational cost The biggest limitation throughout the project was the lack of computational resources necessary to perform text generation tasks on large Norwegian language models. While we tried various QLoRA configurations and `norallm/normistral-11b-warm`, computational limitations led to us using `norallm/normistral-7b-warm` and resulted in restricted fine-tuning. A more in-depth fine-tuning could reduce hallucinations and improve coherence.

Absence of Gold-Standard reference The lack of a gold-standard reference specifically for book reviews, resulted in the inability to calculate automatic evaluation scores like ROUGE. Although human evaluations can provide nuanced insights on the four dimensions, they are costly to scale and more time consuming than automatic evaluation scores.

Prompt engineering Another limitation that affected NorMistral-7b-warm was prompt engineering. The model showed improvements for some titles when we tested the prompt suggested by Samuel et al. (2025) indicating that further work on prompt engineering could result in better generated summaries.

Errors in dataset While exploring the dataset we discovered that some users made incorrect assumptions about the book they were reviewing. This was seen for titles like: 'Inferno' or 'Flink pike', where users reviewed books with these titles but by incorrect authors. This caused the models to generate summaries based on the incorrect summarization input, which does not correspond to the actual book and its author.

6.2 Ethical considerations

The user-generated reviews we used can contain language that reflects or reinforces social bias, discrimination, or misinformation, which could increase the potential of transferring harm in the model-generated summaries. An approach to reduce this risk is to apply filters that exclude reviews with discriminatory language. However, this might cause unintentional censoring of user reviews of sensitive or controversial topics and skew the representation in the dataset.

Our models hallucinate and misstate facts, to address this, we suggest clearly communicating to users of websites like bokelskere.no or Goodreads.com, that the summarized reviews are machine-generated to emphasize the potential limitations. Although our dataset did not require explicit anonymization, real-world deployment should remove such information in the preprocessing step to eliminate personally identifiable information. Finally, automation of review summarization could reduce the diversity of perspectives typically available in full-length reviews written by users. This similarity in reviews could lead to a loss of nuance, particularly in the case of complex or divisive subjects (van der Meer et al., 2024).

7 Conclusion and future work

We developed and evaluated a sentiment-aware abstractive summarization pipeline for Norwegian book reviews, while addressing two research questions: *How can we use abstractive summarization to create Norwegian book review summaries?* and *How do the models Llama-2-7b-chat-norwegian and NorMistral-7b-warm compare in generating Norwegian sentiment-aware abstractive summaries?*

Using a corpus of Norwegian user-generated book reviews, we applied both Norwegian language models to generate summarized reviews of the top 40 most rated books. These were then evaluated through human evaluation using the SummEval framework (Fabbri et al., 2021). Our results revealed that Llama-2-7b-chat-norwegian consistently outperforms NorMistral-7b-warm on three of the evaluation dimensions, confirming that it is more suitable for generating abstractive Norwegian summarizations. We addressed the first research question by describing the complete process, starting with collecting and preprocessing data, followed by finetuning for the summarization task, which resulted in generating summarized book reviews that are readable for Norwegian users. The second question was answered by comparing the results of the human evaluation for both models, which identified that the fine-tuned Llama-2-7b-chat-norwegian model performs better than the NorMistral-7b-warm model.

Future work should include creating a gold-standard dataset that could be used to perform automatic metric evaluation, reducing the time spent on human evaluation, and providing new insights. Furthermore, using a novel dataset, and preprocessing the sentiment tags could enhance the quality of the generated output. Additionally, using larger language models such as NorMistral-11b-warm and llama-2-13b-chat-norwegian could provide more robust generative sentiment-aware summaries.

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